

Theory

A **Genetic Algorithm (GA)** is a population-based **optimization and search technique** inspired by the process of **natural evolution**. It works on a set of possible solutions called a *population* and uses **selection**, **crossover**, and **mutation** operators to evolve better solutions over generations.

Basic Concepts

1. **Chromosome:** A string representing a candidate solution (e.g., binary string 1010011101).
 2. **Gene:** Each position (bit) in the chromosome that carries information about the solution.
 3. **Population:** A group of chromosomes evaluated together in each generation.
 4. **Fitness Function:** Quantifies how good each solution is; higher fitness means a better solution.
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Algorithm Steps

1. **Initialization:** Randomly generate an initial population of chromosomes.
 2. **Fitness Evaluation:** Compute fitness for each chromosome (in One-Max, it is the number of 1s).
 3. **Selection:** Choose parent chromosomes based on fitness probability (e.g., *Roulette Wheel Selection*).
 4. **Crossover (Recombination):** Combine parts of two parents to produce new offspring (e.g., *Single-Point Crossover*).
 5. **Mutation:** Randomly flip bits with a small probability to maintain diversity (*Bit-Flip Mutation*).
 6. **Replacement:** The new population replaces the old one and the process repeats for several generations.
 7. **Termination:** Stop when the best fitness is achieved or a maximum number of generations is reached.
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Mathematical Expression

Each generation (t) evolves towards higher fitness according to:

$$P(t + 1) = \text{Select}(\text{Crossover}(\text{Mutate}(P(t))))$$

Applications

- Function optimization
 - Scheduling and planning
 - Machine learning feature selection
 - Neural network weight tuning
 - Control system parameter optimization
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Advantages

- Works on large, complex, and nonlinear search spaces.
- Does not require derivative information.
- Provides near-optimal solutions through global search.

Disadvantages

- Computationally expensive for large populations.
 - Convergence can be slow.
 - May get trapped in local optima if diversity is lost.
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